

DATABASE SYSTEMS

Chap 5~Database Normalization

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Learning Objectives

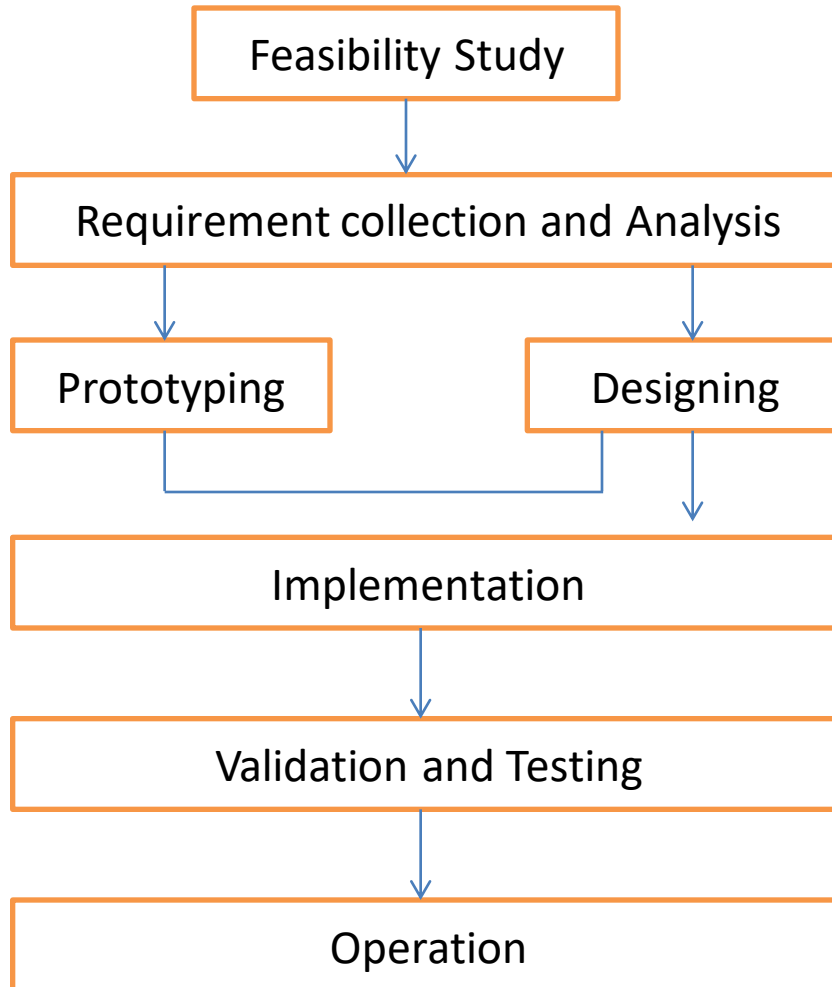
- This deals with various phases in database design, objectives of database design, database design tools.
- The chapter examines the characteristics of a poor table structure and the problems it creates.
- Learn how to design a good table structure and how to repair a poor one.
- Gives concepts on:
 - Various **phases** in database design
 - Database **design tools**
 - Identify **anomalies** in tables
 - **Functional dependency**
 - Concept of **normalization** and **different normal forms**

Introduction

- Database design process integrates relevant data in such a manner that it can be processed through a mechanism for recording the facts.
- The database design is a **complex process**.
- The degree of complexity increases if there are **many-to-many** relationships among individual components.

Introduction

- Steps in database design



Introduction

- Steps in database design

- Feasibility Study

- This is the process of evaluating whether a project or business idea is practical and achievable before implementation.
- The objective of creating the database must be clear, before going further, you must go through the following study:
 - ✓ **Technical feasibility:** Technical feasibility checks whether the required technology, equipment, and technical skills are available to successfully develop and run a project.
 - ✓ **Economical feasibility:** The benefits or profits gained from the project are greater than the costs spent on it.
 - ✓ **Market feasibility:** This checks whether there is demand or acceptance for the product or service in the market. It focuses on: (Customer needs , Competition , Market size , Sales potential , User acceptance)
- E.g. E- Learning platform for secondary schools

Introduction

- Steps in database design
 - Requirement collection and Analysis
 - Decide what data to be stored,
 - How data will be used
 - People who are going to use the database must be interviewed repeatedly
 - Assumptions about the stated relationships between various parts of the data must be questioned again and again.

Introduction

- Steps in database design
 - Prototyping and Design
 - It implies a procedure for analyzing and organizing data into a form suitable to support business requirements and make use of strategic technology
 - It includes conceptual, logical and physical design
 - Implementation
 - It involves development of code for database processing
 - It also involves the installation of new database contents.

Objectives of Databases Design

1. Efficiency

- The design should make full and efficient use of the facilities provided.
- With online database, the users should interact without any time delay

2. Integrity

- The database should be as accurate as possible,

3. Privacy

- Database should not allow unauthorized access to files.

Objectives of Databases Design

4. Security

- Database once loaded, it should be safe from physical corruption whether from hardware or software failure or from unauthorized access
- This is a general requirement of most databases

5. Implementation

- Conceptual model should be simple and effective so that mapping from conceptual model to logical model is easy.

6. Flexibility

- The database should not be implemented in a rigid way that assumes the business will remain constant forever
- Database must be capable of responding readily to changes

Database Design Tools

- Advantages and disadvantages of database design tools

✓ Advantages	Disadvantages
✓ The amount of code to be written is reduced as a result the database design time is reduced	✓ More expenses involved for the tool itself.
✓ Chances of errors because of manual work are reduced	✓ Developers might require special training to use the tool
✓ Easy to convert the business model to working database model	
✓ Easy to ensure that all business requirements are met with	
✓ A higher quality, more accurate product is produced	

Database Design Tools

- Commercial Database Design tools

1. CASE Studio2

- Powerful database modeling, management, and reporting tool
- <http://www.casestudio.com/enu/default.aspx>

2. Design for Databases V3

- Database development tool using an entity relationship diagram
- <http://www.datanamic.com/dezign>

3. DBDesignr4

- Visual database design system that integrates database design, modeling

Database Design Tools

- Commercial Database Design tools

- 5. ER/Studio

- Multilevel data modeling application for logical and physical database design and construction
 - <http://www.embarcadero.com/products.erstudio/index.html>

- 6. Happy Fish Database Designer

- Visual database design tool supporting multiple database platforms.
 - Happy Fish generates complete DDL scripts, defining metadata with table creates, indexes, foreign keys
 - <http://www.embarcadero.com/products/erstudio/index.html>

- 7. Oracle Designer 2000

- Provides complete toolset to model, generate and capture the requirements and design of enterprise applications
 - <http://www.Oracle.com/technology/products/designer/index.html>

Database Design Tools

- Commercial Database Design tools

- 8. Qdesigner

- It is an enterprise modeling and design solution that empowers architects, DBAs, developers, and business analysts to produce IT solutions
 - <http://www.quest.com/QDesigner>

- 9. Power designer

- It offers a modeling solution that analysts, DBAs, can tailor
 - <http://www.sybase.com/products/powerdesigner/>

- 10. xCase

- Database design tools which provides data modeling environment.
 - www.xcase.com

Redundancy and Data Anomaly

- **Redundant data** means storing the same information more than once.
- Redundancy can lead to anomalies.
- The different anomalies are:
 - Insertion anomalies
 - Deletion anomalies
 - Updating anomalies

Redundancy and Data Anomaly

- Given a table

EmpID	EmpName	DeptID	DeptName	DeptLocation
101	Alice	D01	IT	Kigali
102	Bob	D01	IT	Kigali
103	Carol	D02	HR	Nairobi
104	David	D02	HR	Nairobi
105	Emma	D01	IT	Kigali
106	John	D03	Finance	Kampala

- Insertion Anomaly
 - We cannot insert a department without inserting a **member of staff** that works in that department. You cannot insert it unless an employee exists.
- Update Anomaly
 - Change the **Location of IT department** require many updates otherwise it resulting in data inconsistency.
- Deletion Anomaly
 - By removing employee 106, we have removed all information pertaining to the **Finance** department

Redundancy and Data Anomaly

- Repeating Group

- It is an attribute (or set of attributes) that can have more than one value for a primary key value.

Staff no.	Job	Dept. no.	Dept.name	City	Contact number
100	Sales man	10	Sales	Trichy	43545, 45385, 98766
101	Manager	20	Accounts	Coimbatore	67545, -----, -----
102	Clerk	30	Accounts	Chennai	-----, -----, -----
103	Clerk	30	Operations	Chennai	-----, -----, -----

- For each contact number, we have to store the data of the STAFF which leads to more storage space (more memory)
- Repeating groups are not allowed in a relational design, since all attributes have to be **atomic**.

Functional Dependency

- Functional dependencies are the **relationships among the attributes within a relation**
- They provide a formal mechanism to express constraints between attributes.
- If attribute A functionally depends on attribute B, then for every instance of B you will know the respective value of A.
- Notation
 - $A \longrightarrow B$
 - The meaning of this notation is:
 - “B” is functionally dependent on “A”
 - “A” is called determinant, “B” is called object of the determinant.

Functional Dependency

- **E.g. 1**

- Student ID $\longrightarrow$ GPA.
- The meaning is the grade point average (GPA) can be determined if we know the student ID.

Student ID	Name	GPA

- **E.g. 2**

- Child $\longrightarrow$ Mother
- Every child has exactly one mother.
- The attribute mother is functionally dependent on the attribute child
- If we specify a child, there is only one possible value for the mother.

Functional Dependency

- Compound Determinants

- More than one attribute is necessary to determine another attribute in an entity
- **E.g.:** . Grade is made by: Internal mark and external mark
 - . Since more than one attributed is necessary to determine the attribute grade, it is an example of compound determinant.

- Full Functional Dependency

- An attribute is fully functionally dependent on a second attribute if and only if it is functionally dependent on the second attribute but not on any subset of the second attribute
- **E.g.:**
 - The roll number and subject number determines the grade
 - Roll No, Subject Number $\longrightarrow$ Grade

Functional Dependency

- Partial Functional Dependency
 - A non-key attribute depends on only **part of a composite key**, not the whole key.

StudentID	CourseID	StudentName	CourseName

(StudentID, CourseID) : composite key

Partial dependency : StudentID $\rightarrow$ StudentName

Partial dependency : CourseID $\rightarrow$ CourseName

Functional Dependency

- Transitive Dependency

- It exists when there is an intermediate functional dependency
- **Notation**
 - $A \longrightarrow B, B \longrightarrow C$, and if $A \longrightarrow C$ then it can be stated that the transitive dependency exists
 - $A \longrightarrow B \longrightarrow C$
- **Example**

STAFF		
STAFF NUMBER	DESIGNATION	SALARY

- It is to be noted that the staff number determines the designation
- The designation obviously determines the salary

STAFF NUMBER $\longrightarrow$ DESIGNATION
DESIGNATION $\longrightarrow$ SALARY
STAFF NUMBER $\longrightarrow$ SALARY

- There is a transitive dependency between STAFF NUMBER and SALARY

Normalization

- Definition

- It is the process of organizing data in a database.
- It is the analysis of **functional dependencies** between attributes.
- It is the process of decomposing relations with anomalies to produce well-structured relations.

- It includes: creating tables and establishing relationships between those tables according to rules designed both to protect the data and to make the database more flexible by eliminating two factors:

- Redundancy
 - Redundant data wastes disk space and creates maintenance problems
- Inconsistent dependency (Anomalies)

Normalization

- Purpose of Normalization

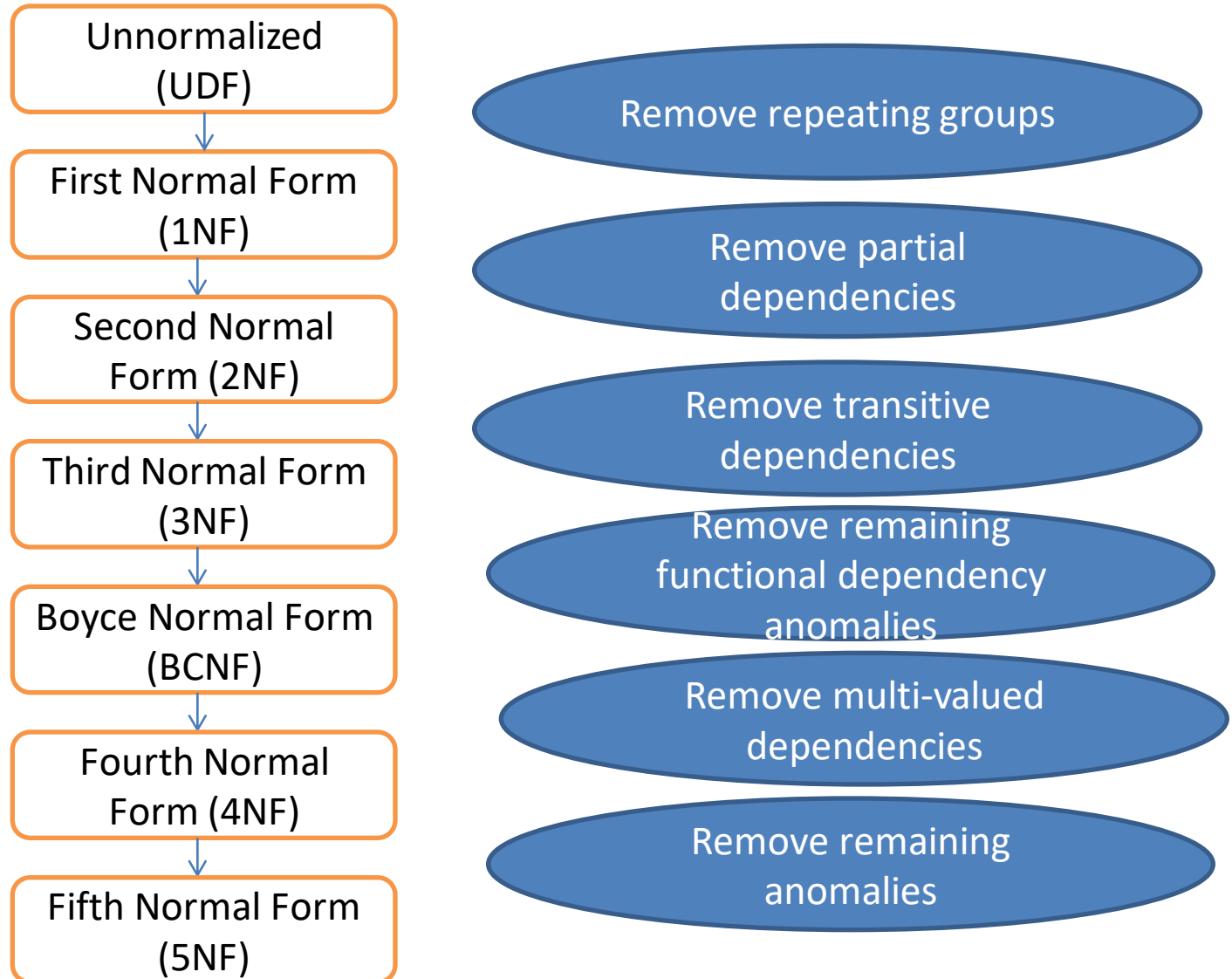
1. To avoid redundancy by storing each fact within the database only once
2. To put data into the form that is more able to accurately accommodate change
3. To avoid certain updating “anomalies”
4. To facilitate the enforcement of data constraint

Normalization

- The degree of normalization is defined by normal forms.
- The normal forms in an increasing level of normalization, are
 - First Normal Form (1NF)
 - Second Normal Form (2NF)
 - Third Normal Form (3NF)
 - Boyce-Codd Normal form (BCNF)
 - Fourth Normal Form (4NF)
 - Fifth Normal Form (5NF)
 - 3NF is considered good enough

Normalization

- Steps in Normalization



Normalization

- First Normal Form (1NF)

- A table is in first normal form if and only if all columns **contain only atomic values**; that is, there are no repeating groups (columns) within a row,
- It is to be noted that all entries in a field must be of same kind and each field must have a unique name, but the order of the field is irrelevant
- Each record must be unique and the order of the rows is irrelevant.

- Second Normal Form (2NF)

- A table is in second normal form if and only if it is in **1NF** and every non-key attribute is **fully dependent on the primary key**

- Third Normal Form (3NF)

- To be in Third Normal Form, the relation must be in **2NF** and **no transitive dependencies** may exist within the relation
- A transitive dependency is when an attribute is indirectly functionally dependent on the key

Example

- Un-normal Form to 1NF

DEPARTMENT (unnormalized form)		
Department number	Department name	Location
1	Nilgiris	{ Coimbatore, Chennai }
2	Subiksha	{ Chennai, Tirunelveli }
3	Krishna	Trichy
4	Kannan	Coimbatore

- Has repeating group
- The values are not atomic

Example

- Solution 1

DEPARTMENT (first normal form)

Department number	Department name	Location1	Location2
1	Nilgiris	Coimbatore	Chennai
2	Subiksha	Chennai	Tirunelveli
3	Krishna	Trichy	
4	Kannan	Coimbatore	

- Divide the column location into location 1, location2, ...
- Some departments will be in one place, in such a case more NULL values will be there in the table

Example

- Solution 2

DEPARTMENT table

Department number	Department name	Location
1	Nilgiris	Coimbatore
1	Nilgiris	Chennai
2	Subiksha	Chennai
2	Subiksha	Tirunelveli
3	Krishna	Trichy
4	Kannan	Coimbatore

- Insert tuples for each location
- The drawback is that there are more repeating values, hence more number of rows

Example

- Solution 3

Department number	Department name	Department number	Location
1	Nilgiris	1	Coimbatore
2	Subiksha	1	Chennai
3	Krishna	2	Chennai
4	Kannan	2	Tirunelveli
		3	Trichy
		4	Coimbatore

- Decompose the relation DEPARTMENT into two tables
- The process of splitting the table into more than one table is called normalization

1NF to 2NF

- Example

Emp_ID	Emp_NAME	Proj_ID	Proj_NAME	Total_time
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- Total time implies the time taken to complete the project
- Total time depends on the nature of the project and the Employee
- Thus total time is determined by the Emp_ID and Proj_ID
- Breaks the relation into three relations EMPLOYEE and HOURS_ASSIGNED, PROJECT
- EMPLOYEE (Emp_ID, Emp_NAME)
- PROJECT (Proj_ID, Proj_NAME)
- HOURS_ASSIGNED (Emp_ID, Proj_ID, TOTALTIME)

Example

- 2NF to 3NF

HOSTEL		
Roll number	Building	Fee
100	main	600
101	additional	500
102	new	650

- Roll number is the key for the table HOSTEL, since the other two columns depends on Roll number, the table is in 2NF
- It is not in 3NF because of transitive dependency
- The table is prone to modification anomalies,
- Removing Roll Number 101, will also deletes the room and building costs.

Example

- 2NF to 3NF (Solution)

Roll number	Building
100	main
101	additional
102	new

building	Fees
main	600
additional	500
new	650

- Avoid the transitive dependency

End

